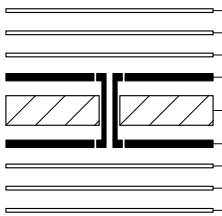
FIC-100A FABRICATION DOCUMENT

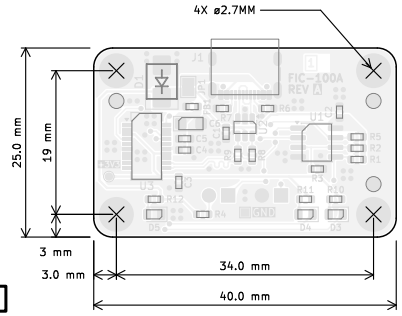
LAYER STACKUP LEGEND



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.49mm	FR4_7628	Dielectric	
Copper	**L2 (Sig, GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

TOP FABRICATION (SCALE 1:1)



IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: ENIG

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

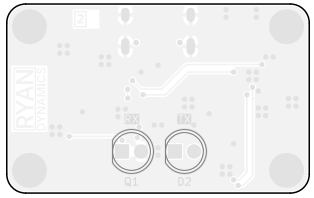
BOARD SIZE	40.000 × 25.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.650 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT USB IR ADPT	Ryan Dynamics		
DRN 2026-04-24 R. HICKS	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
CHK			
ENG APP	FIC-100A		
MFR APP			
SHEET PATH	GIT HASH 62c012b	DRAWING No FABRICATION	A4
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0 / SHEET 1 OF 8	

FIC-100A FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)



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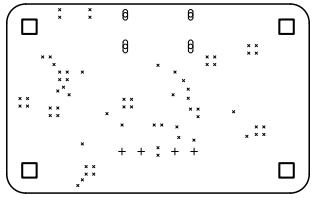
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0 / SHEET 2 OF 8	

FIC-100A FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	66	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
○	4	0.60mm (23.62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
+	4	0.90mm (35.43mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	4	2.70mm (106.30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
	Total 78					

DRILL DRAWING L1 – L2 (SCALE 1:1)



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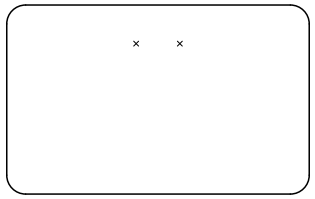
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0 SHEET 3 OF 8	

FIC-100A FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0.65mm (25.59mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total 2					

DRILL DRAWING L1 – L2 (SCALE 1:1)



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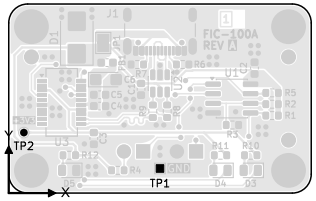
ENG APP	FIC-100A		
MFR APP			
SHEET PATH	GIT HASH 62c012b	DRAWING No FABRICATION	A4
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0	SHEET 4 OF 8

FIC-100A FABRICATION DOCUMENT

TOP TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	GND	20.00	3.25
TP2	+3V3	2.00	8.00

Ref.	Net	X [mm]	Y [mm]



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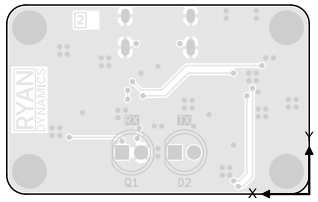
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0 / SHEET 5 OF 8	

FIC-100A FABRICATION DOCUMENT

BOTTOM TEST POINTS (SCALE 1:1)

Ref.	Net	X [mm]	Y [mm]



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PROJECT	USB IR ADPT
DRN	2026-04-24 R. HICKS
CHK	
ENG APP	
MFR APP	

Ryan Dynamics

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SHEET PATH	GIT HASH 62c012b	DRAWING No FABRICATION	A4
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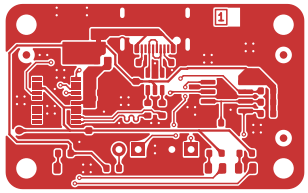
FIC-100A FABRICATION DOCUMENT

L1 (Sig, PWR) (SCALE 1:1)



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MFR APP			
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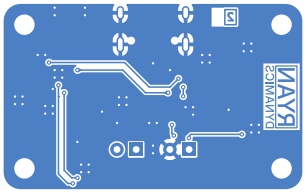
FIC-100A FABRICATION DOCUMENT

L2 (Sig, GND) (SCALE 1:1)



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ENG APP	FIC-100A		
MFR APP			
SHEET PATH	GIT HASH 62c012b	DRAWING No FABRICATION	A4
FILENAME FIC-100A.kicad_pcb	VARIANT RELEASED	REVISION 0.1.0 / SHEET 8 OF 8	